

PHAN HUU KHA

Aspiring Backend Engineer

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PROFILE

Software Engineering Student at Sai Gon University with a strong passion for Backend Development and Systems Internals. Gained hands-on experience through building personal projects with event-driven messaging (Kafka), in-memory caching (Redis), and concurrency control. Eager to learn, tackle real-world engineering challenges, and contribute as a Backend Engineering Intern.

EDUCATION

Sai Gon University

Engineer's Degree in Software Engineering

Sept. 2023 – June 2028

Current GPA: 3.4 / 4.0

TECHNICAL SKILLS

Languages: Java (17/21 LTS), SQL, JavaScript (ES6+), Bash Shell

Backend Frameworks: Spring Boot 3.5, Spring Security 6 (JWT / OAuth2 / RBAC), Spring Data JPA/Hibernate, WebSocket (STOMP), SSE, RESTful APIs

Databases & Queues: PostgreSQL, MongoDB, Redis Stack (RedisBloom / Pub/Sub), Apache Kafka (KRaft / Avro)

DevOps & Infrastructure: Docker & Docker Compose, Google Jib, Jenkins CI/CD, GitHub Actions, Tailscale (Funnel), Nginx, Antigravity CLI

Testing & Architecture: JUnit 5, Mockito, MockMvc, Swagger / OpenAPI 3.0, Event-Driven Architecture (EDA), Distributed Clustering

PROJECTS

ChatWeb – Distributed Messaging Platform

🌐 github.com/phwkha/ChatWeb-RealTime

Role: Backend Developer & System Designer | Java 21, Spring Boot 3.5, Kafka, Redis, PostgreSQL, MongoDB, STOMP, Nginx, Docker Compose

- Engineered a Write-Behind pipeline via **Kafka (KRaft, Avro)** with dual consumers (sub-10ms push & 200-msg batch to MongoDB) to decouple real-time delivery from disk I/O bottlenecks and isolate write failures with DLT.
- Routed clustered WebSocket traffic using a **Redis Hash** session directory and node-specific Pub/Sub channels to scale horizontally across server replicas without broadcast storms.
- Protected DB resources using **RedisBloom Cuckoo Filters** for O(1) account preflight, atomic Redis Lua rate limiters, a 5s ZSet anti-flapping queue, and MongoDB `$max` to prevent write amplification.
- Secured edge ingress via **Nginx** with private Root CA upstream TLS, paired with **Spring Security 6** (JWT, OAuth2) and atomic PostgreSQL `token.version` to enforce instant multi-device session revocation.
- Orchestrated a 10+ container topology via **Docker Compose** with healthcheck chains and 10 named volumes to eliminate startup race conditions and guarantee isolated, persistent services.
- Built automated CI/CD via **GitHub Actions** and **Google Jib** (daemonless layered builds to Docker Hub) with **Swagger / OpenAPI 3.0** to standardize delivery and streamline interactive API testing.

Confession-Wall – Homelab & Community Platform

🌐 github.com/phwkha/Confession-Wall

Role: Fullstack Engineer (Personal Lab) | Spring Boot, Linux, Tailscale

🌐 kha-dev.impala-tritone.ts.net

- Provisioned a physical **Ubuntu bare-metal server** hosting containerized workloads under explicit Docker resource quotas to achieve 100% self-hosted operational control at zero cloud expense.
- Engineered Zero-Trust public ingress via **Tailscale Funnel** by tunneling HTTPS over WireGuard to Nginx, bypassing residential CGNAT and eliminating 100% of open-port attack vectors without port forwarding.
- Automated a 5-stage containerized **Jenkins CI/CD pipeline** (`/var/run/docker.sock`) with H2 testing, multi-stage builds, and rolling updates with curl healthchecks to cut deploy time to <3 minutes with auto-rollback.
- Synchronized real-time **Server-Sent Events (SSE)** streaming with Spring `afterCommit` transaction hooks and unbuffered Nginx to deliver sub-15ms pushes while eliminating ghost broadcasts on rollbacks.

LANGUAGES

Vietnamese: Native | **English:** Professional working proficiency (technical documentation, specifications)